

LLD Case Study 4: Car Rental System (Hertz / Zoomcar)

Target Patterns: Factory Pattern, Strategy Pattern, State Pattern

Key Engineering Focus: Multi-branch vehicle inventory, dynamic pricing algorithms, vehicle lifecycle states, and reservation management.

1. Problem Statement & Functional Requirements

Design an automated Car Rental Management platform operating across multiple cities and airport branches.

Requirements:

- 1. **Vehicle Hierarchy (Factory Pattern):** Manage diverse vehicle classes (Economy Hatchback, Compact SUV, Luxury Sedan, Electric Vehicle) with distinct specifications and rates.
- 2. **Dynamic Pricing (Strategy Pattern):** Calculate rental costs using swappable pricing algorithms (Base Daily Rate, Surge/Peak Season Pricing, Distance/Mileage Model).
- 3. **Reservation State Machine (State Pattern):** Reservations progress from Pending → Confirmed → PickedUp → Completed (or Cancelled).
- 4. **Multi-Location Inventory:** Allow vehicle pickup at one branch and return at a different branch with inter-branch inventory rebalancing.

2. Architecture & UML Class Diagram



3. Production-Grade Python Implementation

